The Cyclistic 2024 Membership Growth Challenge

Cyclistic Case Study 2024

Portfolio: Joey's Analytics Portfolio
Kaggle version: Coming Soon
Case Study Analysis: Case Study - A - Cyclistic
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How Do Cyclistic Riders Differ?

Key Business Question:

- How do annual **members** and **casual** riders use Cyclistic bikes differently?

Casuals	Members
Purchase rides individually, less frequent riders	Enroll in an annual plan

Strategic Challenge:

- What key differences in bike usage between casual riders and members can Cyclistic uncover to overcome barriers and drive annual membership conversions?

At a Glance

- Used R-Studio for all analysis and data visualization
- Analyzed ride start time, end time, and duration by rider type
- Data excludes personally identifiable info, limiting individual-level insight
- Gaps in data: no demographic or payment method info
- Analysis focuses on overall behavioral trends between member types

Prepare Overview

- Verified column types (integer, character, datetime) were correct
- Removed duplicates, empty rows, and irrelevant data
- Filtered out extreme ride length outliers
- Used summary statistics and visual checks to confirm data integrity
- Tools: RStudio, shared via Kaggle and GitHub

Process Overview

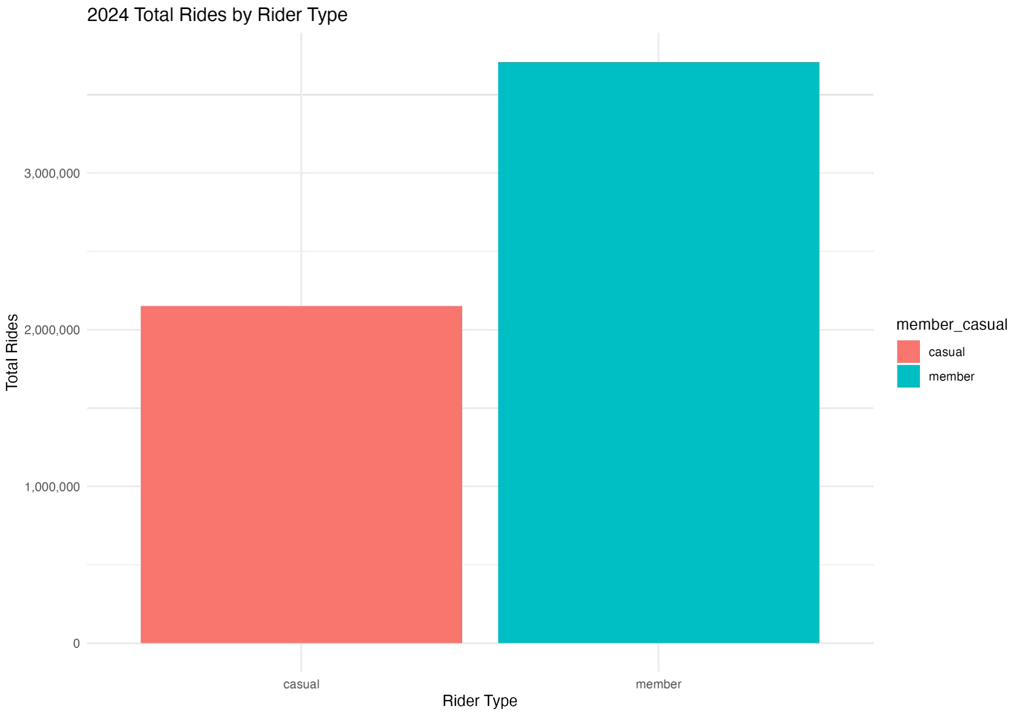
- Performed error checks on time entries and missing values
- Used progressive dataframes for each cleaning step
- Summarized data by day, month, and rider type
- Validated summaries before moving to analysis
- Documented steps and transformations in R

Cyclistic at a Glance (2024 Overview)

- Snapshot of key rider activity metrics from January to December 2024.
- Helps us understand when, how often, and who is using Cyclistic bikes the most.
- Highlights usage trends like busiest hour (5 PM), top month (September), and popular bike type (electric).
- Establishes a baseline for identifying member vs. casual rider behavior differences.
- Informs where marketing strategies could best target casual riders for conversion.

1	Total Number of Rides	5,860,568
2	Average Ride Length by Rider Type	**Casual**: 25.15 **Member**: 12.77
3	Busiest Time of Day	5:00 PM
4	Busiest Day	Saturday
5	Busiest Month	September
6	Busient Season	Summer
7	Most Popular Bike	Electric
8	Most Rides by	Members

2024 Total Rides by Rider Type



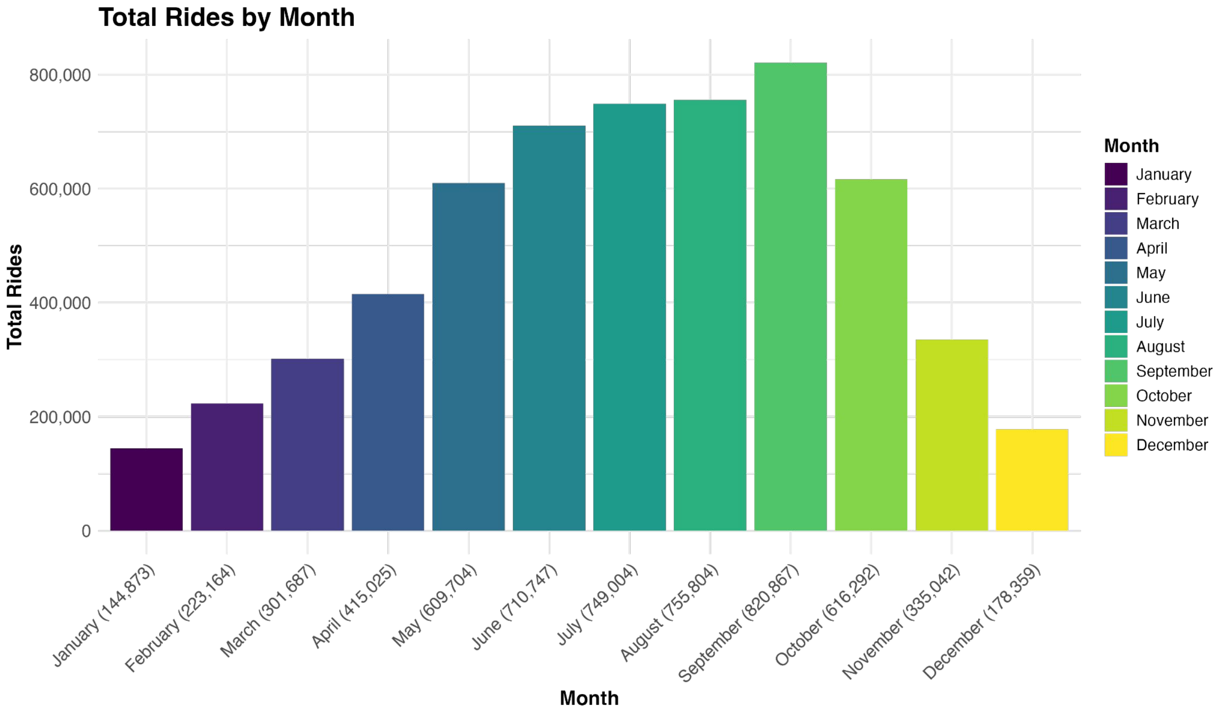
Members took the majority of rides, suggesting stronger engagement than casual users.

- Visual comparison of total rides taken by casual vs. member riders
- Members consistently contributed the most rides
- Opportunity exists to shift casual users into higher-usage membership tier

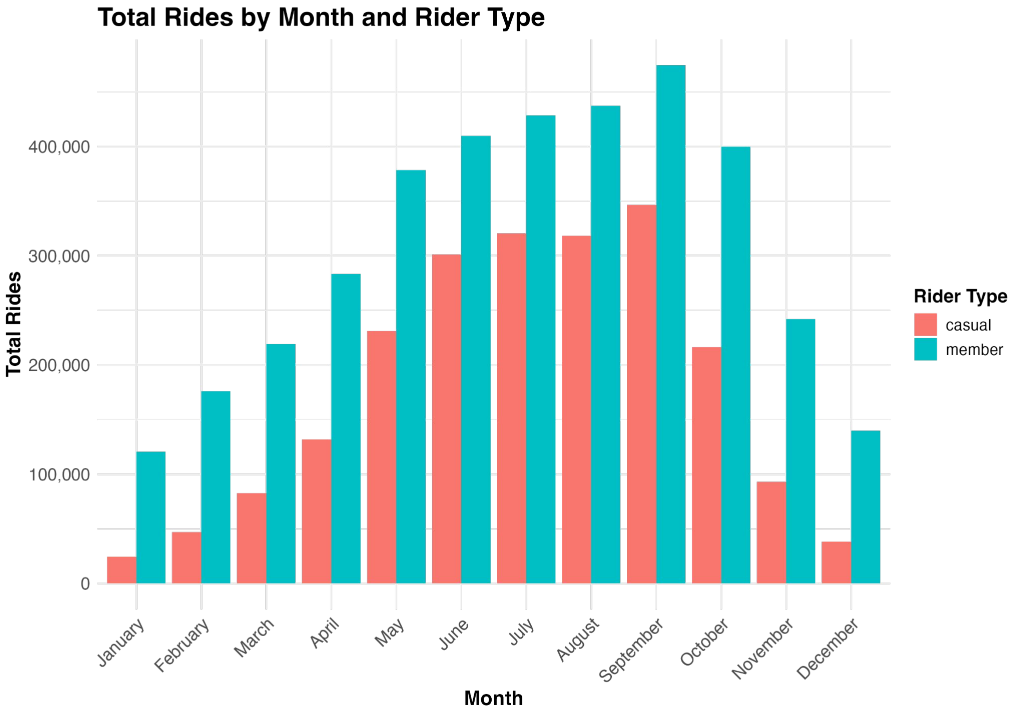
Total Rides by Month

Ride volume clearly aligns with seasonal weather changes, peaking in summer.

- Seasonal patterns emerge strongly in the data
- Peak ride months: Summer, especially June through September
- Sharp drop in colder months



Who's Riding When? Monthly Breakdown by Rider Type



Casual riders are mostly active in summer, while members ride year-round.

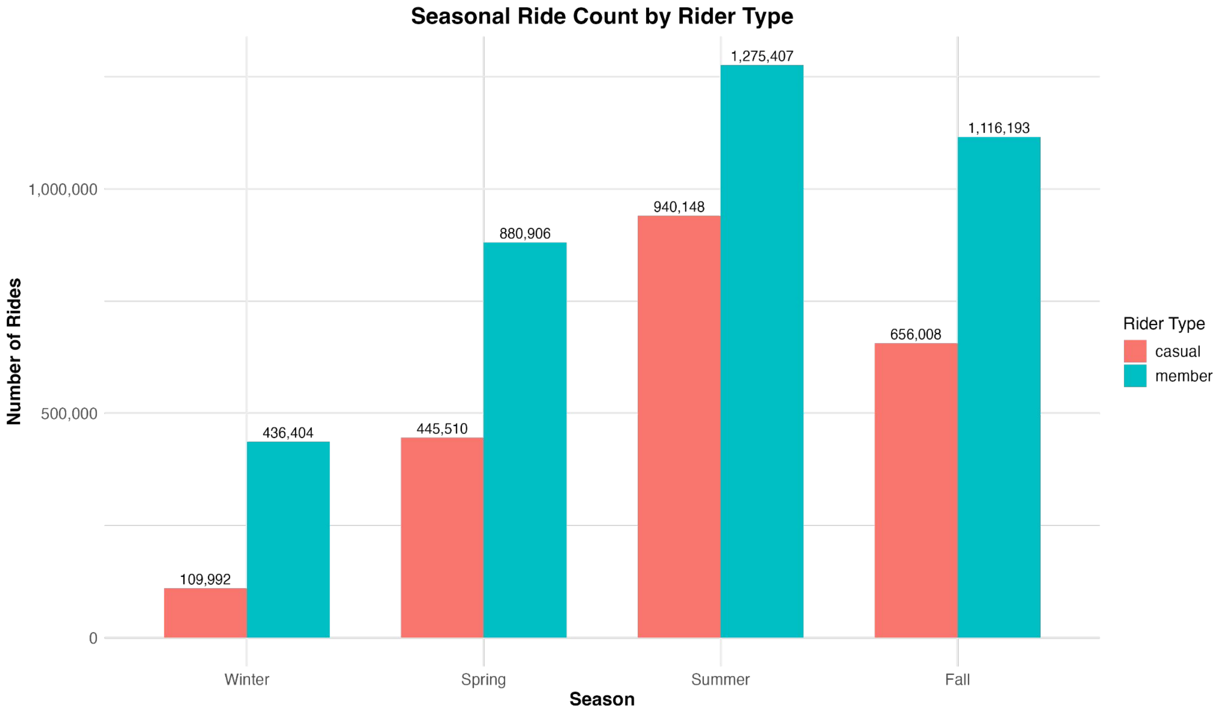
The busiest season for both types of customer was the summer,

- Members maintain steady usage beyond peak season
  - Member numbers remained high through October
- Indicates casuals are seasonal or recreational riders
- Casual riders were high in June, July, August, and September only

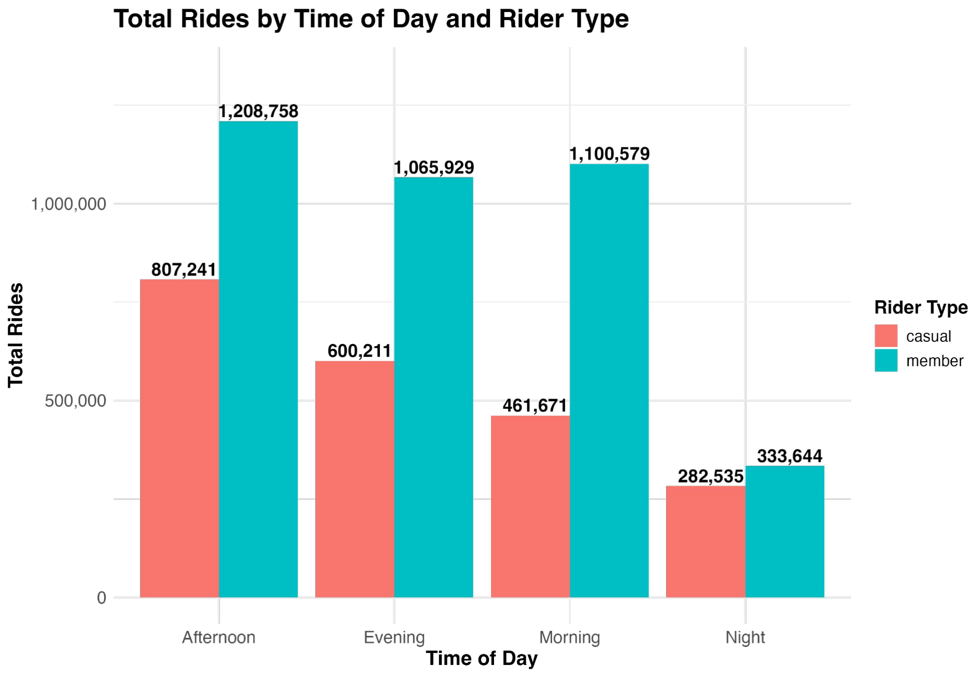
Who's Riding When? Seasonal Breakdown by Rider Type

Summer dominates for all users, but member activity is more balanced throughout the year.

- Summer is the busiest season for both groups
- Casual usage heavily concentrated in summer
- Members show moderate engagement even in winter



Total Rides by Time of Day and Rider Type



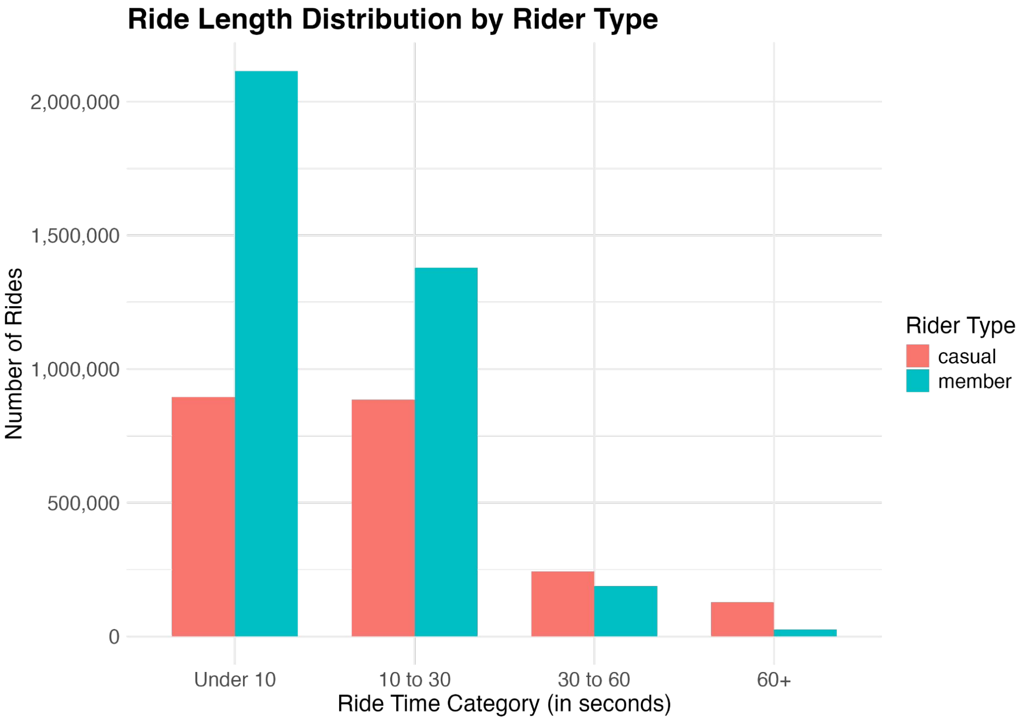
This bar chart shows how ride activity is distributed by time of day for both rider types.

- Displays total rides by hour of day for each rider type
- Members and casuals both ride more in late afternoon
- Morning hours are less popular for casual riders
- Helps identify peak demand hours by group

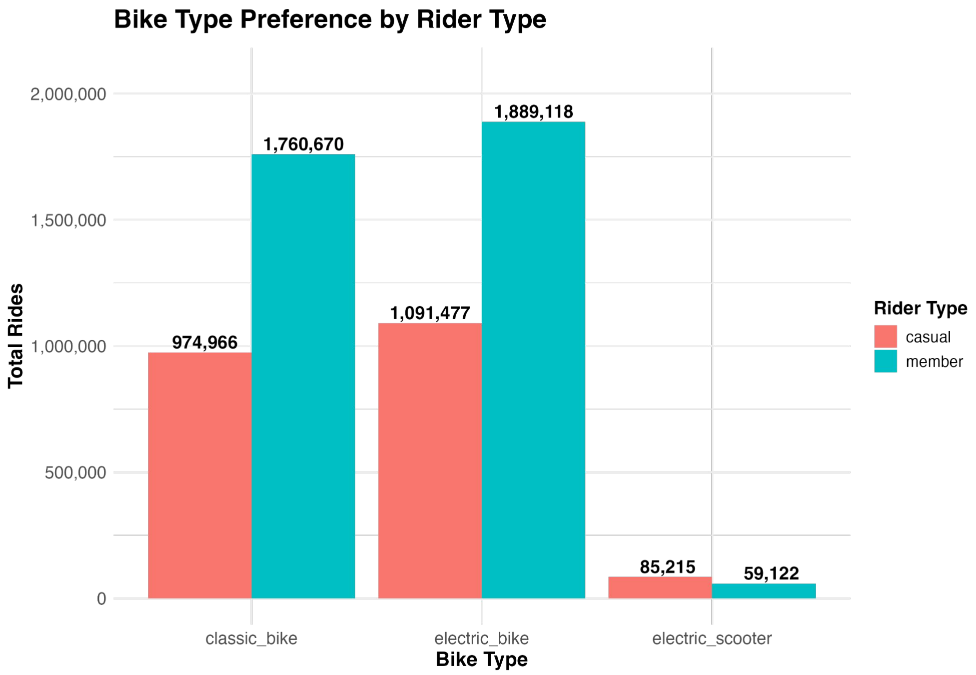
Distribution of Total Rides by Time of Day and Rider Type

This chart shows how ride durations differ between casual and member riders.

- Casual riders are more likely to take longer trips
- Member rides cluster under 30 minutes, indicating faster trips
- Duration reflects differing motivations: leisure vs. utility



Bike Type Preference by Rider Type



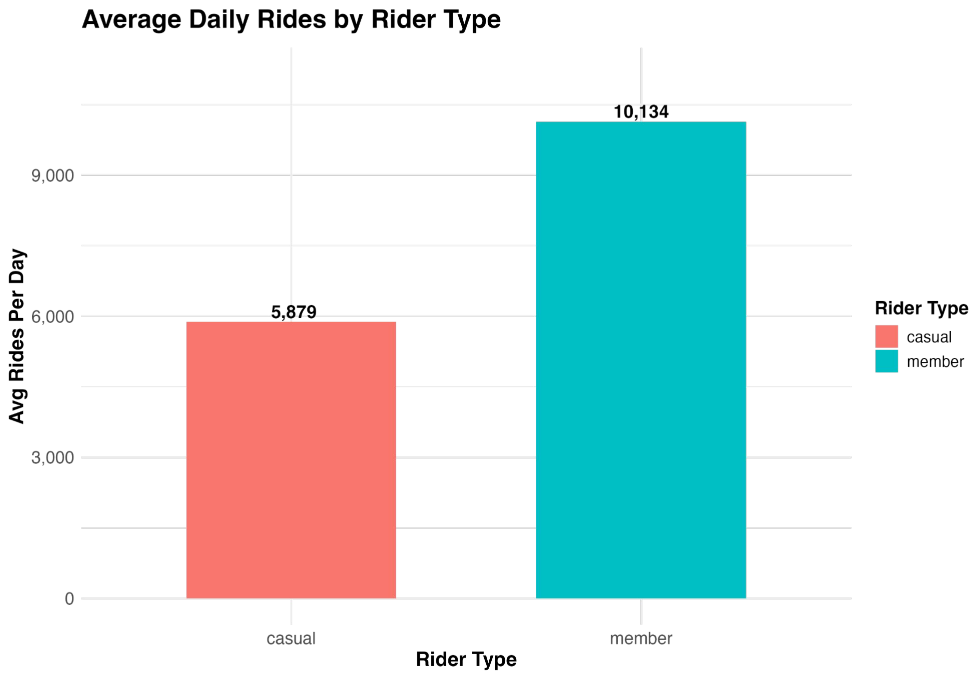
Electric bikes are most popular overall, especially with casual riders.

- Electric bikes were most popular overall
- Casual riders used electric and docked bikes more often
- Members showed higher use of classic bikes

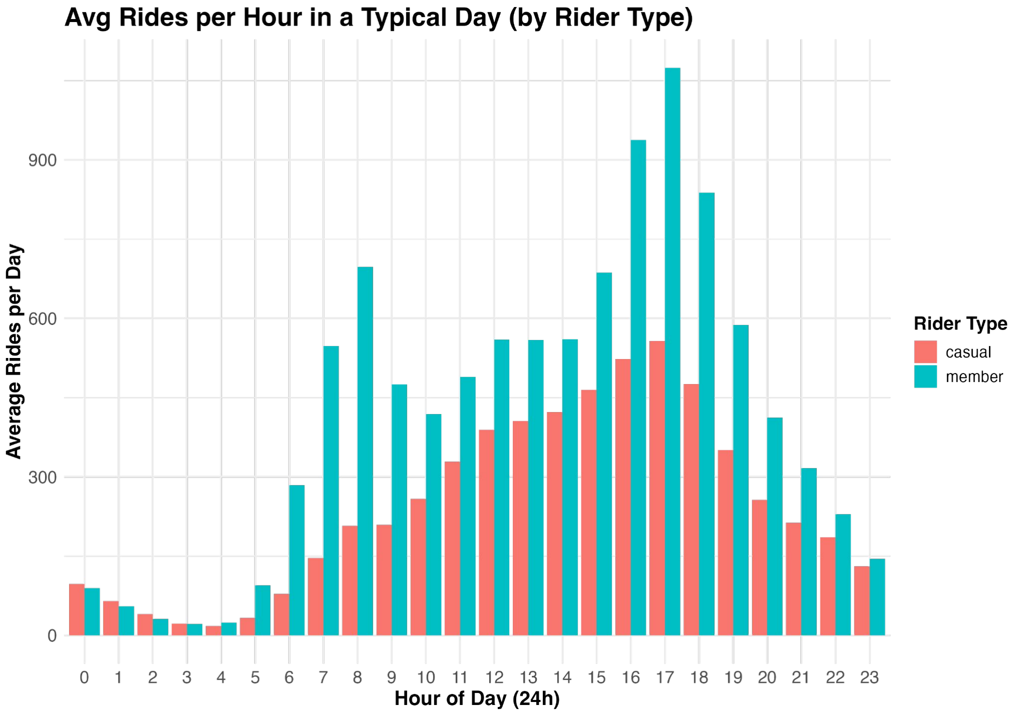
Average Daily Rides by Rider Type

Members ride more frequently, reflecting higher engagement and retention.

- Members average more rides per day
- Casuals ride less frequently but spike during summer
- Engagement higher among members overall



Daily Ride Rhythm: Members vs. Casuals (by Rider Type)



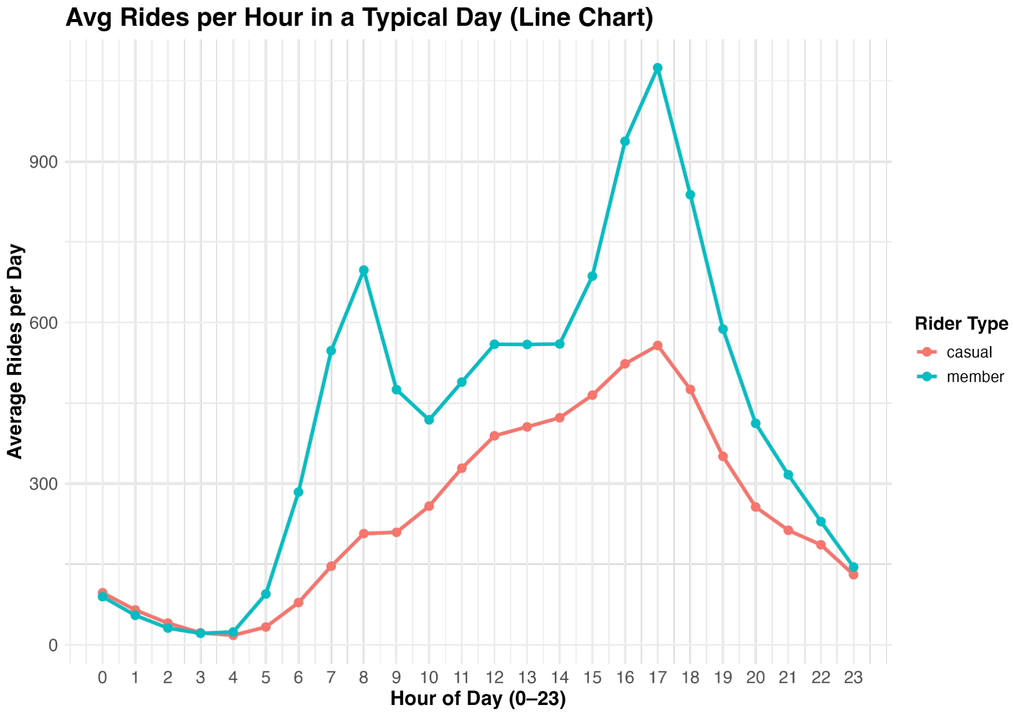
Members ride early and late, casuals prefer the afternoon.

- Busiest hour for both groups is 5 PM
- Casual riders also ride in early afternoon
- Member rides start earlier, peaking at commute times

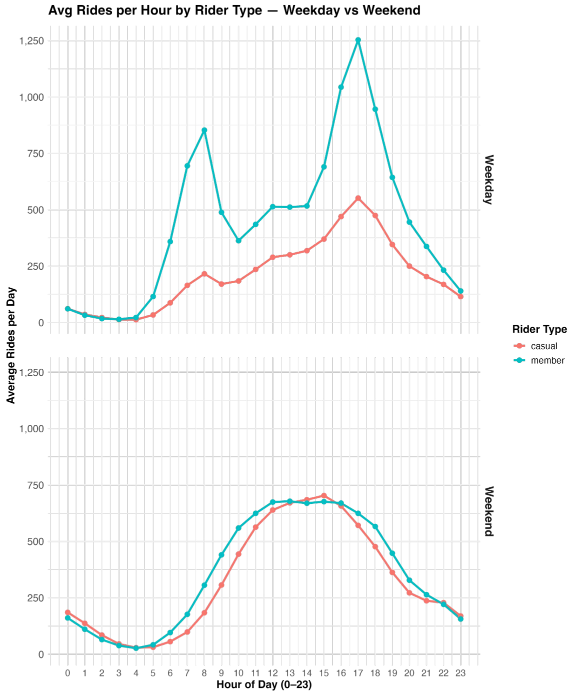
Avg Rides per Hour in a Typical Day (Line Chart)

This chart clearly shows two daily usage patterns based on rider type.

- Visualizes rider behavior throughout the day
- Casual rides rise slowly and peak later
- Member rides peak twice — morning and evening



Avg Rides per Hour by Rider Type - Weekday vs Weekend



Members ride more during weekdays, casuals ride more on weekends.

- Weekday: Member rides peak during commute hours
- Weekend: Casual riders dominate in the afternoon
- Clear behavioral shift by day type

Summary - Key Differences Between Member and Casual Riders

The analysis of 2024 ride data reveals distinct behavioral patterns between Cyclistic's annual members and casual riders:

- Members show more consistent usage across the year, averaging 10,134 rides per day, whereas casuals ride less frequently with 5,879 per day. This supports the idea that converting casual riders into members can increase overall ride engagement.
- **Ride Timing:** Members consistently ride during weekday morning and evening commute hours, peaking around 8 AM and 5–6 PM. In contrast, casual riders prefer riding later in the day, with afternoon and early evening peaks, particularly on weekends. This suggests a strong recreational trend among casual users, while members rely on Cyclistic for commuting.
- **Ride Frequency:** Although casual riders contribute to seasonal spikes in usage (especially summer months), annual members show more consistent and frequent engagement year-round. Members average more daily rides compared to casual riders, reflecting greater retention and habitual usage.
- **Bike Preference:** Classic bikes are favored by both groups, but casual riders show a higher relative usage of docked and electric bikes, possibly drawn to novelty or convenience.

Recommendations

The final analysis suggests clear opportunities to tailor marketing and service strategies toward converting casual riders into long-term members.

1. Promote Annual Memberships Around Peak Casual Usage Periods (Summer and Weekends)

- Casual rider activity peaks in warmer months and on weekends.
- Launch targeted membership campaigns during these high-traffic periods to convert engaged riders while interest is high.

2. Highlight Commuter Benefits in Digital Ads

- Members show consistent weekday commuting behavior.
- Position annual membership as a cost-effective and reliable commuting alternative, particularly for urban professionals.

3. Feature Popular Bikes in Campaigns (Especially Electric and Docked Bikes)

- Casual riders are more likely to use docked and electric bikes. Marketing that features these options prominently, along with membership benefits may increase conversion.

Conclusion - Marketing Insights

Recreational vs. Commuter Personas

- The behavioral divide between members and casual riders suggests two distinct personas.
- Use personalized content one targeting weekend leisure riders, another targeting weekday commuters.

Electric Bike Appeal

- Casual riders may be more drawn to electric and docked bikes for convenience or novelty.
- Highlight unlimited access to these options in your membership offerings.

Leverage Usage Habits for Retargeting

- Data shows casual riders cluster around specific times (weekends, afternoon hours).
- Use this to time push notifications, email reminders, or social ads offering special membership deals.

Next Steps for Driving Membership Growth

To build on these insights and maximize conversions, Cyclistic should explore the following targeted actions:

- Run A/B tests with summer-targeted membership promotions for casual riders
- Conduct follow-up surveys to identify barriers preventing casuals from converting
- Explore pricing bundles and incentives for high-frequency casual users
- Integrate ride behavior data into CRM for personalized retargeting campaigns